1: Department of Meteorology, University of Reading | 2: Met Office Hadley Centre |
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S2S Predictability of North Sea Marine Heatwaves: A Process-Based Approach

Do SEAS5 ensemble members capture North Sea MHWs for the right reasons?

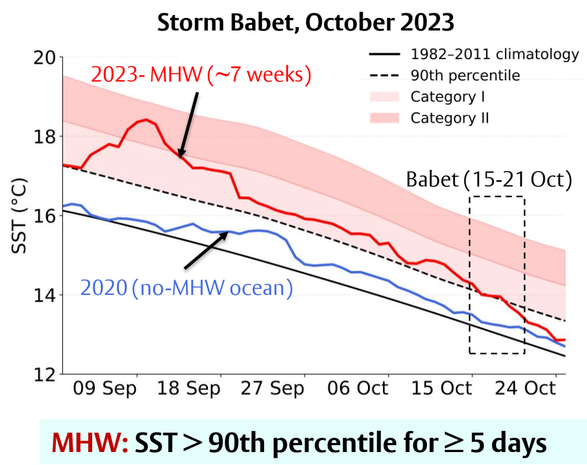
Piyali Goswami[1] | Ted Shepherd[1] | Ségolène Berthou[2] | Magdalena Balmaseda[1,3] | Regina Rodrigues[4]

Why does predicting marine heatwaves matter?

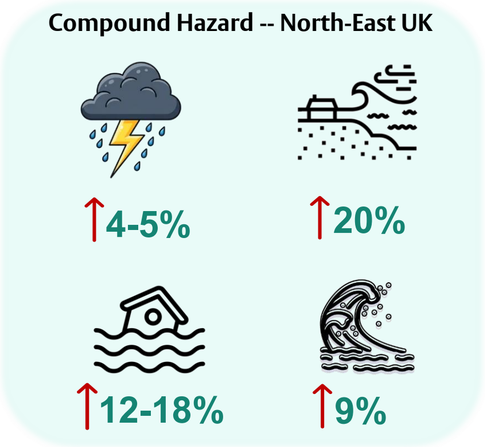
Marine heatwaves (MHWs) are periods of unusually warm sea-surface temperatures that as well as their direct marine impacts, can influence air-sea interactions and extreme weather impacts.

The North Sea is a shallow shelf sea, so its heat content is strongly influenced by air-sea exchange.

During **Storm Babet (October 2023)**, a pre-existing North Sea MHW amplified rainfall, river discharge, storm surge and wave power (Goswami et al., 2026).



MHW: SST > 90th percentile for ≥ 5 days



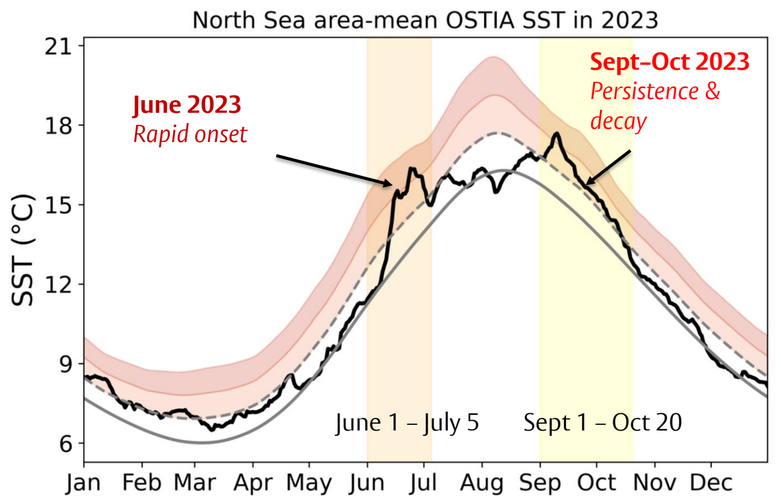
Event-based S2S Predictability: June vs September–October 2023

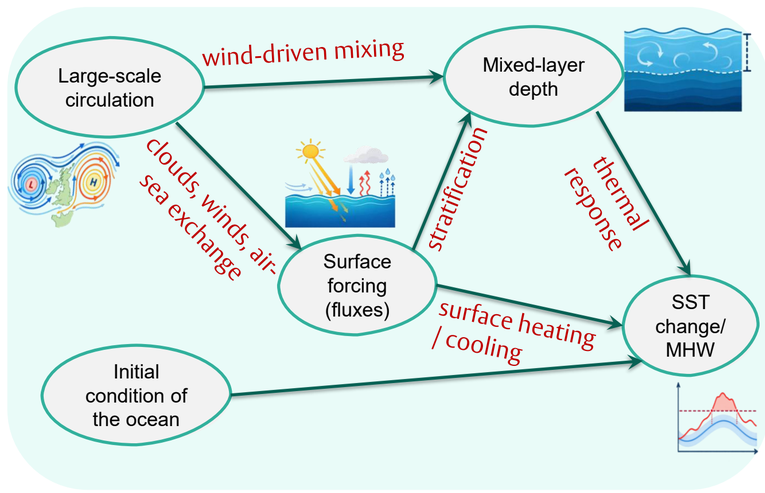
Two contrasting North Sea MHWs in 2023: **June** and **September–October**.

51-member ECMWF SEAS5 ensembles

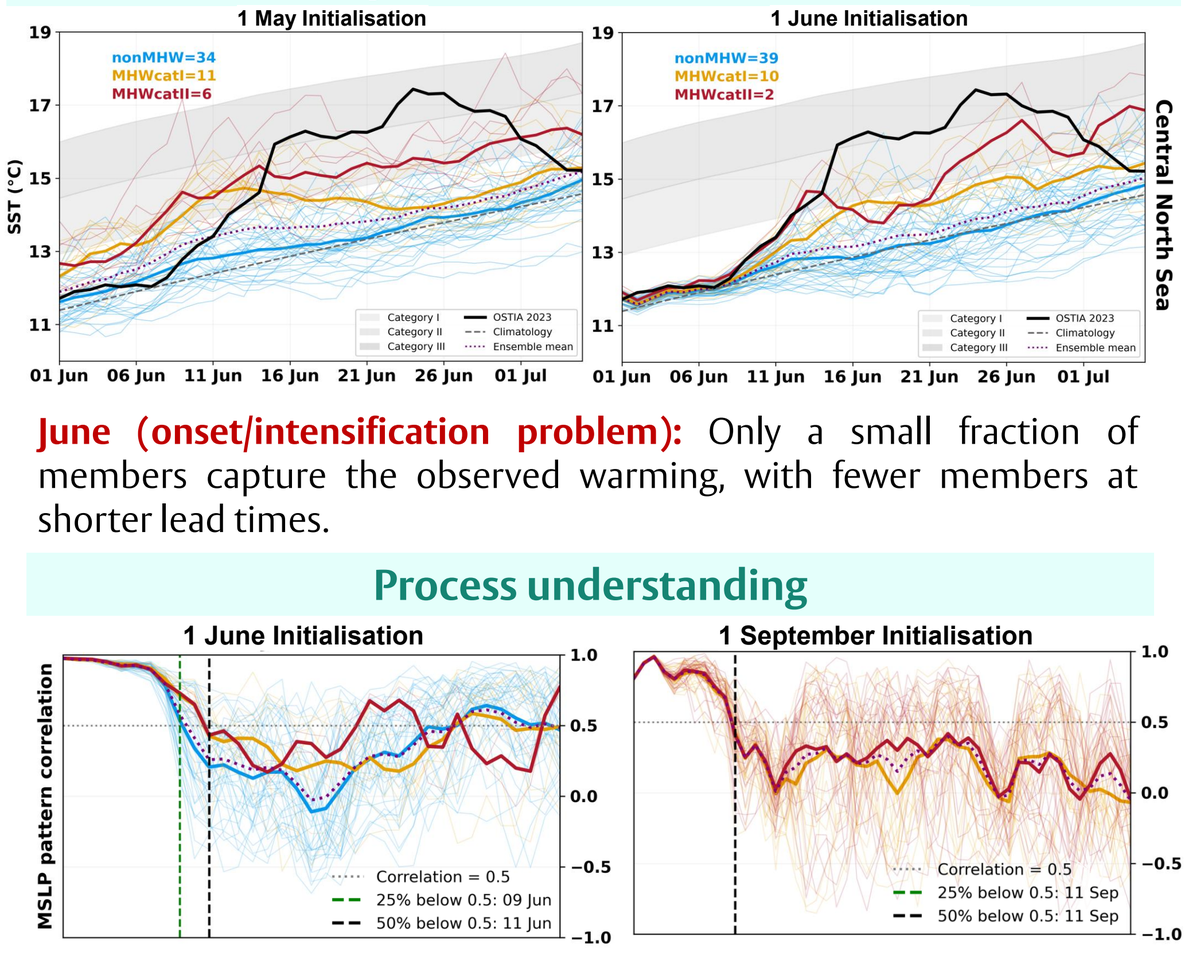
- nonMHW : predict no MHWs
- MHW Cat I : >= 5 Cat I days & <3 Cat II days
- MHW Cat II : >= 3 Cat II days

Physical diagnostics: MSLP pattern correlation, cumulative net surface heat flux and mixed-layer depth are compared across the forecast groups



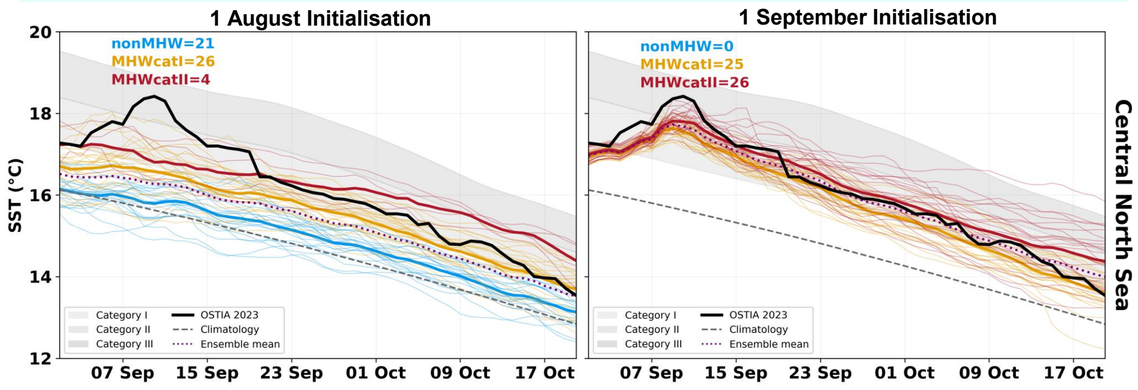


June Event SST Forecast



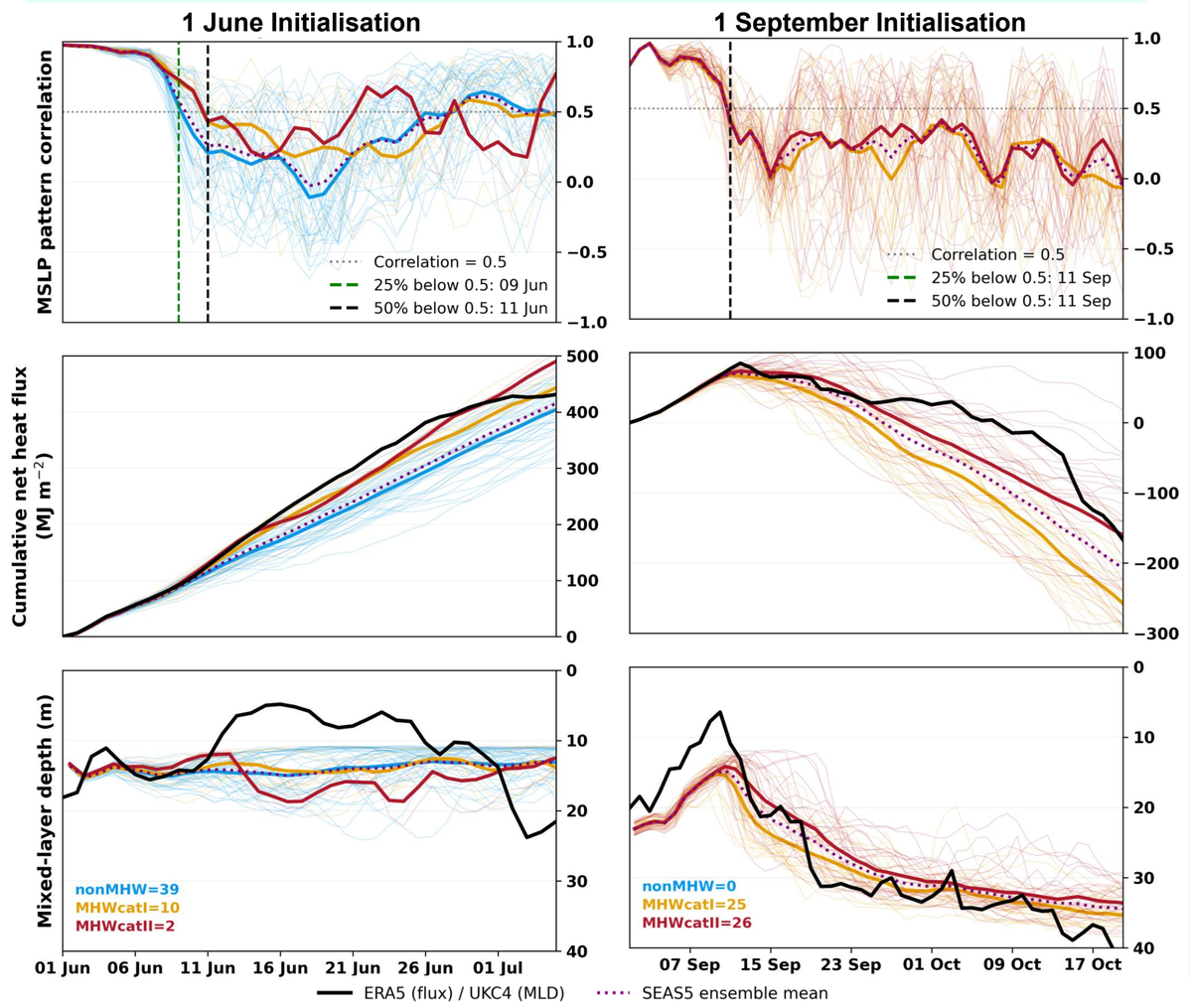
June (onset/intensification problem): Only a small fraction of members capture the observed warming, with fewer members at shorter lead times.

September-October Event SST Forecast



Sept–Oct (persistence/decay problem): by September initialisation, all members produce MHW, but many of them do not capture the decay.

Process understanding



What controls the predictability?

Contrasting predictability: The two events highlight different forecast challenges: capturing **the rapid June onset** and the **Sept-Oct decay**.

Atmospheric circulation: MSLP pattern correlations deteriorate within **~10 days** in both events, with earlier divergence in June.

Air–sea heat exchange: June experiences sustained positive surface heat flux, while Sept-Oct shows **net heat loss despite persistent warm SST forecasts.**

Ocean mixing: Differences in mixed-layer evolution between global SEAS5 and regional UKC4 highlight potential **limitations in representing the ocean's thermal response**.

Preconditioning: Higher initial OHC anomalies in Sept suggest that **ocean thermal memory** may contribute to predictability. *(not shown)*

Looking at SST alone is not enough: Examining the underlying processes helps explain the forecast behaviour.

(Manuscript in preparation)

Future work

Extend the causal framework to more MHW events and investigate how **long-term climate warming and climatological baseline choices** influence MHW identification and assessed S2S predictability.

Why does predicting these MHWs matter?

Goswami, P., Berthou, S., Shepherd, T. G., Volonté, A., Mahmood, S., Castillo, J. M., ... & Balmaseda, M. A. (2026). Marine heatwave amplifies extreme multi-hazards of extratropical cyclone Babet. *Natural Hazards and Earth System Sciences*, *26*(7), 3213-3230.

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